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Unraveling the demographic and socioeconomic factors shaping subjective health status in Chile over three decades: implications for health policy

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Abstract

Background Subjective health status (SHS) is a key indicator of population health and is influenced by various demographic and socioeconomic factors. However, there is a limited understanding of how these factors have shaped SHS in Chile over the past three decades and how health policies have addressed these determinants. This study aims to analyze the predictors of SHS at an individual level and evaluate the alignment of health-focused social programs with these predictors to promote equitable health outcomes in Chile.

Method We utilized data from the World Values Survey (WVS) to examine the impact of demographic and socioeconomic factors on SHS in Chilean adults from 1990 to 2019, applying two probit regression models with and without time control. Additionally, we analyzed the scope of 479 health-focused social programs implemented by the Chilean Government between 2012 and 2021 to determine their contribution to improving SHS and other life domains such as work, education, and social protection.

Results Age, female, primary education, retirement, and lower-income quintiles are associated with lower SHS over time. Conversely, being a parent, having higher education, and having higher income quintiles relate to higher SHS. Education and income consistently shape SHS over time. Health-focused programs interact with other areas, providing social protection and education. Nevertheless, there are opportunities to positively affect other life domains under an intersectional social policy perspective.

Conclusions The findings underscore the potential of well-informed health policies to substantially improve SHS. They suggest that a holistic approach, including environmental and social determinants, is crucial for advancing public health outcomes in Chile. We offer some recommendations for policy purposes.

Keywords Self-reported health status, Demographic factors, Socioeconomic factors, Health policy

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Introduction

According to the World Health Organization (WHO, 1946), "health is not merely the absence of disease or infirmity, but a state of complete physical, mental, and social well-being". This comprehensive definition underscores the importance of addressing social determinants of health. However, there is a knowledge gap regarding how these determinants influence SHS, particularly in the Chilean context, where socioeconomic and demographic disparities can significantly shape health perceptions [1].

SHS has gained substantial traction due to its robust predictive validity for mortality, particularly among older adults, even when controlling for chronic and psychological diseases [2, 3]. Despite this, gaps remain in understanding how demographic and socioeconomic factors interact to influence SHS, especially in countries like Chile, where structural inequalities may play a critical role. These gaps highlight the need for integrated approaches that take into account the unique characteristics of Latin America.

The most recent findings of the Chilean National Socioeconomic Characterization Survey revealed that in 2017, the majority of the population (59.7%) rated their health status between 6 and 7 on a 7-point scale, indicating a generally positive perception [4]. However, these statistics do not capture the underlying inequalities or how factors such as sex, education, and income shape these perceptions. This lack of detailed information limits the design of social policies that effectively address health disparities.

This study aims to evaluate how demographic and socioeconomic variables influence SHS in the adult population of Chile. In addition, it seeks to identify gaps in current social policies and propose evidence-based strategies to address health inequalities. By addressing these questions, this work contributes to filling a critical void in the literature on health in Chile.

Why subjective health status matters?

SHS is a self-reported health perception measure that has gained prominence as a reliable predictor of mortality and morbidity [5, 6]. Its importance lies in its ability to capture individual experiences and perceptions of health, offering a more holistic perspective beyond clinical indicators. SHS reflects the intersection of physical, mental and social well-being, according to the definition of health by the World Health Organization as a state of complete well-being [1]. This multidimensional understanding is particularly relevant in the Latin American context, where social inequalities profoundly shape health outcomes [7].

Research has consistently shown that SHS is influenced by demographic factors such as age and sex, as well as socioeconomic determinants such as income and education [6, 8]. For example, studies indicate that SHS tends to decline with age and is generally lower among women, highlighting the role of life course dynamics and gendered health disparities [5]. Education emerges as a critical determinant, with higher levels associated with better health perceptions due to increased access to resources, social support, and health literacy [7]. Similarly, income plays a pivotal role, as individuals in higher income quintiles consistently report better SHS compared to those in lower quintiles, underscoring the structural nature of economic inequities [6].

In Latin America, the study of SHS has revealed complex interactions between social determinants of health and cultural contexts. Large-scale studies such as the Latin American Study of Nutrition and Health (ELANS), the Brazilian Longitudinal Study of Adult Health (ELSA-Brasil), and the SALURBAL project provide key insights into the region's structural and contextual factors shaping SHS.

The research from the ELANS study demonstrated significant disparities in SHS across socioeconomic strata. Individuals with a higher socioeconomic status (SES) consistently reported better health perceptions related to improved access to nutrition, healthcare, and educational opportunities. These findings emphasize the role of structural inequities in shaping SHS and highlight the compounded disadvantages faced by individuals in lower SES groups [6]. Similarly, ELSA-Brasil data identified poor SHS as strongly associated with low income, limited education, and high occupational stress. This underscores how socioeconomic disadvantage operates through multiple pathways to affect health perceptions [8].

The SALURBAL project analyzed self-reported health data from over 100 Latin American cities and revealed that urban environments and city-level socioeconomic conditions amplify individual-level disparities. Sex also emerged as a significant modifier, with women reporting poorer SHS compared to men, particularly in cities with more significant socioeconomic inequalities [6]. These studies collectively illustrate the multidimensional nature of SHS and the need to consider structural determinants alongside individual characteristics.

Despite significant progress, the study of SHS in Latin America is characterized by gaps in integrating findings into sociopolitical frameworks. Many studies focus on descriptive analyses or isolated determinants, limiting their ability to inform policies to reduce health inequalities comprehensively. Understanding how demographic and socioeconomic factors shape SHS provides a foundation for designing targeted interventions that address health inequities. By integrating SHS data with

policy planning, governments can prioritize resources for the most vulnerable groups and create programs that improve well-being in multiple dimensions. This approach is particularly crucial in the Chilean context given the persistent inequalities in healthcare access, education quality, and income distribution in the country [9, 10].

Case setting

Chile's unique socio-geopolitical landscape makes it an exemplary case for analyzing SHS within the broader demographic and socioeconomic disparities context. Located in South America, Chile is home to approximately 19 million people and is often regarded as one of the most economically developed nations in the region. However, this perceived development is accompanied by deeply entrenched inequalities that significantly impact health outcomes, including SHS [11].

One of the most prominent features of Chile's socioeconomic structure is its high level of income inequality, reflected in a Gini index of 44.9 in 2021, one of the highest in Latin America [11].This level of inequality is further exacerbated by a segmented health system comprising a public sector (FONASA) that serves about 78% of the population and a private sector (ISAPREs) catering to higher-income groups [9]. This dual system reinforces disparities in access to and quality healthcare services, directly affecting health perceptions among socioeconomic groups.

Additionally, based on individual capitalization, Chile's pension system has left many retirees, particularly those in lower-income quintiles, with insufficient financial resources to meet basic needs, including healthcare [10]. Similarly, the public education system, characterized by chronic underfunding, limits opportunities for upward mobility and perpetuates cycles of inequality [12]. These structural vulnerabilities intersect with demographic factors, such as age and sex, to create distinct patterns in SHS.

Geographically, Chile's urban-rural divide further complicates access to healthcare and other social services. Although urban centers like Santiago offer greater access to high-quality services, but rural areas often lack the infrastructure to support equitable health outcomes. This disparity highlights the multidimensional nature of health inequities in Chile [13].

Chile's economic model, rooted in neoliberal reforms introduced in the late 20th century, has also played a critical role in shaping social policies and health outcomes. These reforms prioritized market efficiency over social equity, leading to socioeconomic stratification that is reflected in SHS disparities. For instance, individuals in higher income quintiles consistently report better SHS compared to those in lower quintiles, illustrating how structural inequities permeate individual health perceptions [5].

This study situates SHS within Chile's socio-political framework to explore how demographic and socioeconomic factors shape health perceptions. By analyzing trends over three decades, this research provides a comprehensive understanding of how systemic inequalities influence SHS and offers insights into the unique challenges faced by the Chilean population. These findings have implications for designing targeted policies aimed at reducing health disparities and promoting well-being across all social groups.

Materials and methods

Data and participants

This quantitative and explanatory research exclusively uses secondary data extracted from the World Values Survey (WVS) conducted by the Gallup Corporation from 1981 to 2020. The WVS collects nationally representative samples from almost 100 countries, addressing various topics related to economic development, democracy, religion, gender equality, social capital, and well-being [14]. Currently, the WVS offers social and political researchers the opportunity to understand how people's beliefs, values, and motivations vary over time. The data are available through the corporation's official website (www.worldvaluesurvey.org).

The information in the WVS is collected through a face-to-face questionnaire applied to heads of households or via telephone in hard-to-reach areas. The minimum sample accepted per country for each period is 1,000 cases, which corresponds to a representative subset of the national population aged 18 and over. The cases included in the databases do not admit missing data. For the Chilean case, the survey has a total of 6,700 cases distributed over six collection periods: from 1990 to 1993 ($n = 1,500$), from 1994 to 1998 ($n = 1,000$), from 1999 to 2004 ($n = 1,200$), from 2005 to 2009 ($n = 1,000$), from 2010 to 2014 ($n = 1,000$), and from 2017 to 2019 ($n = 1,000$). It should be noted that the specific years in which the survey was applied in Chile were 1990, 1996, 2000, 2006, 2012, and 2018, respectively. This research uses the six periods to compare variations in subjective well-being in the Chilean population over time.

We also observed how health-focused programs interact with other target life dimensions. For doing so, all programs report which other life domains they address through their outcomes, declaring yes or no through five dimensions: work, housing and environment, social protection, education, and social cohesion.

We processed all program reports freely available on the official social program website as PDF documents

(https://programassociales.ministeriodesarrollosocial.gob.cl).

We built a database with 479 observations corresponding to annual reports from 2012 to 2021. We support data in an Excel datafile for processing descriptive analysis and basic significance tests to observe differences between SHS and sociodemographic variables.

Variable description

The WVS incorporates a variety of themes. Mainly, health status is measured through a Likert-type question systematically asked in the six data collection periods for the Chilean case: In general, how do you evaluate your health status? (1 "Very good", 2 "Good", 3 "Regular", 4 "Poor", 5 "Very poor"). Additionally, a set of personal data from the respondents was used, allowing them to be characterized in sociodemographic and socioeconomic terms to evaluate their effects as independent variables on SHS: Age (in years), respondent's sex ("Female" 0, "Male"), marital status (1 "Married", 2 "Cohabitant", 3 "Divorced", 4 ""Separated", 5 "Widowed", 6 "Single"), parenthood (1 "I am a parent", 0 "I am not"), ethnicity (1 "Yes, I belong to an ethnic group", 0 "No, I do not belong"), head of household (1 "Yes, I am head of household", 0 "No, I am not"), highest educational level (1 "No education", 2 "Basic education", 3 "Secondary education", 4 "Higher education"), occupational category (1 "Full-time worker", 2 "Part-time worker", 3 "Self-employed", 4 "Retired or pensioned", 5 "Houseowner", 6 "Student", 7 "Unemployed"). Income decile in which the respondent is classified (I to X) and period (1 "1990–1993", 2 "1994–1998", 3 "1999–2004", 4 "2005–2009", 5 "2010–2014", 6 "2017–2019").

The health-focused programs database includes information about their scopes of action: citizen participation and organization; alcohol/drugs; education; social protection; science, technology, and connectivity; productive development; independent work and entrepreneurship; housing; culture/arts; judicial, environment and natural resources; urbanism; public spaces and transportation; dependent work; citizen security; and physical and recreational activity. Each one is a dummy variable. Data also incorporates dummy variables about the program's contribution to reducing income poverty, multidimensional poverty, health poverty, education poverty, work and social security poverty, and social cohesion and networks poverty. Finally, data also explore if the programs collaborate to promote each SDGs: affordable and clean energy, decent work and economic growth, industry, innovation, and infrastructure, climate action, life below water, life on land, partnerships for the goals, clean water and sanitation, reduced inequalities, responsible consumption and production, quality education, no poverty, peace, justice, and strong institutions, gender equality, sustainable cities and communities, zero hunger and, good health and well-being.

Data analysis

Two probit regression models were carried out to assess the effect of demographic and socioeconomic variables on perceived health status, both with and without time control. Health status was treated as a dependent variable and coded as a dummy variable, assuming that 1 represents those who report good or very good health and 0 represents those with a regular, poor, or very poor perception of their health. This decision was made because, in ordinal regression models, there was a high loss of cases due to the absence of data in the response categories "very good", "poor", and "very poor". The same treatment was applied to the independent variables, where all of them were transformed into dummies, except for age, which was treated as a continuous variable measured in years.

We addressed probit regression because it is particularly advantageous in modeling binary outcomes like our dichotomous SHS variable. This method aligns with our data's binary nature of health perception, enabling a precise analysis of the probability of positive health reporting. Moreover, the probit model's assumption of normally distributed error terms suits our demographic and socioeconomic variables predicting SHS. Probit regression methods usually allow us to handle large-scale survey data, ensuring robust and reliable statistical analysis and enhancing our findings' methodological rigor and relevance, particularly in informing public health policy in Chile [15].

We also analyzed frequency distributions to observe health status trends in greater detail according to primarily explanatory demographic and socioeconomic variables. The statistical test result is reported to assess whether the differences are statistically significant. Regarding the health programs database, we applied basic descriptive statistics.

Ethics considerations

This study is based solely on publicly available secondary data from the World Values Survey website and the Social Programs Database from the Social Development Ministry in Chile. All data used in this study are anonymous and non-identifiable and were not collected specifically for this study. Therefore, ethical approval was not required for this analysis as it did not involve the participation of human subjects or the collection of primary data. Data confidentiality was ensured following the policies of the World Values Survey.

Results

The results are based on a sample of 6,700 cases. Of these, 52.2% are women and 47.8% men. The average age is 44, with a minimum of 18, a maximum of 99, and a standard deviation of 16 years. Regarding the participants' marital status, 60.5% report living with a partner, while 24% are single, 5.6% are divorced or separated, and 6.3% are widowed. The data also indicate that 75.7% of the participants are parents with one or more children. Regarding the highest educational level achieved, 11.7% obtained a university degree, 31% completed secondary education, and 13.4% did not finish primary education. Regarding occupational characteristics, 37.7% declare themselves full-time workers, 22.7% homemakers, 10.7% entrepreneurs, 9.4% retired, 7.6% students, and 6.5% part-time workers. Finally, considering total family monthly income, 28.8% declare themselves part of the second income quintile, 27.4% of the third, 23.4% of the first, 15.5% of the fourth, and 5.4% of the fifth income quintile.

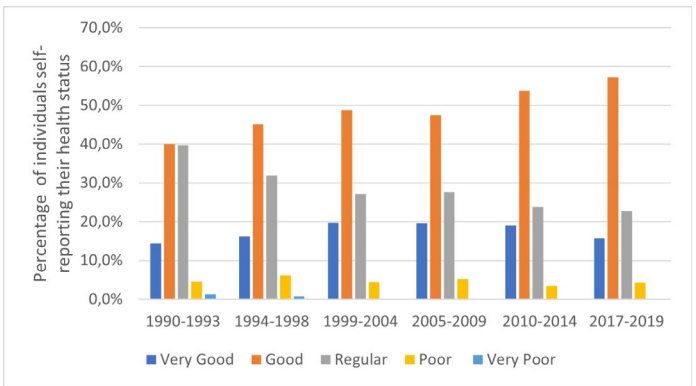
A global analysis of subjective health status indicates that nearly 50% ($n = 3219$) of people evaluate their health as "good", followed by 29.6% ($n = 1979$) who consider it "regular", 17.3% (n=1155) who categorize it as "very good", 4.6% ($n = 310$) who declare a "poor" health status, and 0.4% ($n = 28$) who consider it "very poor". An analysis of the health status perceived by the Chilean population over time suggests that it is generally positive and has systematically improved since 2005 (Fig. 1).

The aggregated results presented above provide an overview of health perceptions within the Chilean population. However, this level of aggregation may obscure critical variations in how different demographic and socioeconomic factors influence personal health perceptions. To address this limitation, we implemented two probit regression models, one including time control and the other without it. These models allowed us to isolate the specific effects of individual demographic and socioeconomic characteristics on self-reported health status, treated as a binary dependent variable.

Table 1 presents the outcomes of these models, which incorporate identical variables except for the time control included in the second model. By comparing the two approaches, we aim to uncover temporal trends while maintaining focus on the core determinants of health perceptions.

In the first model, it is observed that the subjective perception of health status is negatively affected by demographic attributes such as being female and being older. Having only completed primary education and not accessing higher educational levels also negatively impacts the self-assessment of health status. In socioeconomic terms, it was found that performing specific occupational categories, such as being self-employed, retired, or being a house owner or homemaker, negatively affects perceived health status, as does belonging to the lower income quintiles. Conversely, achieving secondary or higher education levels and being part of the top two income quintiles positively affect health status.

In the second model, when controlling for the effect of time, all variables maintain similar impacts on subjective



Note: N = 6691. Mean subjective health status significantly differs across time periods ($F = 19.337, p < 0.001, df = 5, 6685$). 95% confidence intervals for the mean health status by time period are as follows: 1990-1993 (2.37, 95% CI [2.33, 2.41]), 1994-1998 (2.30, 95% CI [2.25, 2.35]), 1999-2004 (2.16, 95% CI [2.12, 2.20]), 2005-2009 (2.19, 95% CI [2.14, 2.23]), 2010-2014 (2.12, 95% CI [2.07, 2.16]), and 2017-2019 (2.16, 95% CI [2.11, 2.21]).

Fig. 1 Percentage of individuals self-reporting their health status over different time periods. Note: *N* = 6691. Mean subjective health status significantly differs across time periods ($F = 19.337, p < 0.001, df = 5, 6685$). 95% confidence intervals for the mean health status by time period are as follows: 1990–1993 (2.37, 95% CI [2.33, 2.41]), 1994–1998 (2.30, 95% CI [2.25, 2.35]), 1999–2004 (2.16, 95% CI [2.12, 2.20]), 2005–2009 (2.19, 95% CI [2.14, 2.23]), 2010–2014 (2.12, 95% CI [2.07, 2.16]), and 2017–2019 (2.16, 95% CI [2.11, 2.21])

Table 1 Regression models assessing the effect of demographic and socioeconomic variables oneself-reported health status

Variable	Model 1	Model 2
Head of Household (HOH)	−0.013	−0.004
	(0.011)	(0.011)
Ethnicity (ETHNIC)	0.015	0.002
	(0.014)	(0.014)
Being a Parent (PARENT)	0.042	0.053*
	(0.028)	(0.028)
Being Female (FEMALE)	−0.078**	−0.109***
	(0.040)	(0.040)
Age (AGE)	−0.009***	−0.012***
	(0.001)	(0.001)
Being Married or Cohabiting (MARRIED)	0.012	0.020
	(0.030)	(0.030)
Being Widowed (WIDOWED)	−0.049	0.002
	(0.050)	(0.050)
Being Divorced (DIVORCED)	−0.002	−0.027
	(0.047)	(0.047)
Having Completed Basic Education (BASIC_ED)	−0.346***	−0.329***
	(0.054)	(0.054)
Having Completed Secondary Education (SECONDARY_ED)	0.109***	0.091***
	(0.035)	(0.035)
Having Completed Higher Education (HIGHER_ED)	0.251***	0.245***
	(0.047)	(0.048)
Being a Full-Time Worker (FULL_TIME)	0.081	0.061
	(0.077)	(0.078)
Being a Part-Time Worker (PART_TIME)	−0.036	−0.023
	(0.097)	(0.098)
Being Self-Employed (SELF_EMP)	−0.181**	−0.105
	(0.088)	(0.089)
Being Retired or Pensioned (RETIRED)	−0.235**	−0.178*
	(0.103)	(0.104)
Being a House Owner or Homemaker (HOMEMAKER)	−0.222***	−0.126
	(0.085)	(0.086)
Being Unemployed (UNEMPLOYED)	−0.137	−0.111
	(0.100)	(0.101)
Belonging to the First Income Quintile (QI)	−0.227***	−0.210***
	(0.036)	(0.036)
Belonging to the Second Income Quintile (QII)	−0.148***	−0.196***
	(0.033)	(0.033)
Belonging to the Fourth Income Quintile (QIV)	0.131***	0.134***
	(0.041)	(0.041)
Belonging to the Fifth Income Quintile (QV)	0.240***	0.273***
	(0.059)	(0.059)
Period (TIME)		0.110***
		(0.011)
_cons	1.083***	0.723***
(0.086)	(0.092)	
N	6699	6699

Source: World Values Survey, 1990–2019. All variables in the models have at least five observations - probit regression. Health status is a dummy variable. Standard errors in parentheses

*$p < 0.10$, **$p < 0.05$, ***$p < 0.01$

health status, except for the occupational categories of self-employed and house owner or homemaker, which lose statistical significance. Time also appears to significantly and positively impact perceived health status.

A descriptive analysis of those variables that significantly impact health status allows deeper insight into some results. Regarding the sex variable, it is interesting that men self-assess their health status better than women. While about 70% ($n = 2208$) of men consider their health status as "very good" or "good", 61.3% ($n = 2166$) of women evaluate their health favorably. In contrast, 6.1% ($n = 215$) of women consider their health to be "poor" or "very poor" compared to 3.9% ($n = 123$) in men.

When incorporating the distinction by age group, it is observed that men's and women's positive self-assessment of their health status decreases with age. However, there are specific nuances after age 60 (Fig. 2). The perception of health status shows no significant differences between men and women between the ages of 61 and 70. However, men between the ages of 71 and 80 have a better perception of health than women, and women aged 81 and older have a better than men.

Regarding the impact of education, the higher the educational level achieved, the better the perception of personal health status (Fig. 3). This trend is systematic as educational levels increase, such that more years in the educational system correspond to a more favorable self-assessment of health status compared to those who did not study or only completed primary education.

A similar trend is observed when examining the perceived health status by income level (Fig. 4). In this case, Whereas quintiles simplify the model, deciles allow a graphical representation to capture finer variations across the socioeconomic spectrum. As we observe, the concentration of people who evaluate their health as poor or very poor tends to decrease systematically as income increases.

There is also a specific polarization of the population in the lower three deciles, where there are people who favorably evaluate their health but also a significant percentage, close to 40%, who consider it "fair". In contrast, the upper three deciles represent 75% or more of people who declare they are in very good or good health.

Health-focused programs and SHS predictors

Chile is committed to social policies to combat poverty and expand access to social services. Over the years, a notable expansion in the scope of social sectors covered by these policies has been observed, with these sectors increasingly recognized as subjects of rights by the Chilean state. Social protection is central to this policy shift, which has become a primary focus for addressing the needs of the country's most impoverished and

vulnerable populations over the past decade. This emphasis has broadened in the last five years to encompass more extensive coverage and benefits grounded in a rights-based approach [16, 17].

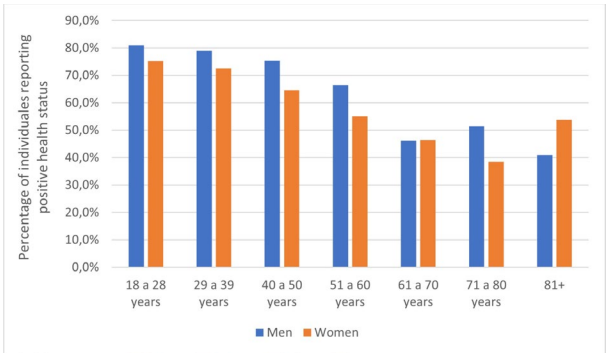
From 2012 to 2021, the Chilean Government implemented an average of 359 social programs annually, ranging from a minimum of 274 programs in 2014 to a maximum of 489 in 2021. A consistent subset of 79 programs was executed annually, with 17 programs persisting for nine years, 45 for eight years, and 64 for seven years. In contrast, 192 programs were maintained for only two years, with a surge to 273 programs in 2021. Predominantly, these programs addressed fundamental needs such as health, education, housing, and social security, with a core set remaining constant over time, while others adapted to specific contexts and local demands. Notably, programs focused on "health and well-being" ranked third after "education" and "social protection", indicating their significant role in Chilean social policy.

Nevertheless, health and well-being programs implemented between 2012 and 2021 fluctuated significantly (Fig. 5). After increasing from 31 to 92 programs between 2012 and 2013, a drastic fall was reported in 2014 with 21 programs. In 2015–2016, the programs implemented increased, followed by decreases in the next two periods.

We observed how the national health focus programs are related to some SHS predictors highlighted in the previous models (Fig. 6). The results illustrate the distribution of health-focused programs across various scopes of action. The X-axis represents specific sectors or areas addressed by these programs, including social protection, education, and alcohol and drug initiatives, which show the highest levels of integration. The Y-axis indicates the number of programs within each sector. Notably, areas such as social participation, sports, and urbanism have a lower representation, highlighting potential gaps in program coverage. This underscores the need for broader integration across underrepresented sectors to enhance the impact of health policies.

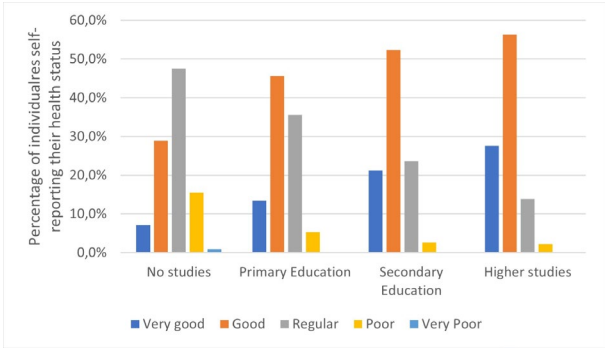
Health-focused programs associated with social protection are particularly impactful, enhancing psychological well-being and reducing financial stress-both known to influence SHS. Additionally, health programs integrated with education promote health in schools through nutritional education and first aid training, equipping communities with essential knowledge for disease prevention and healthy living. Although health programs focused on alcohol and drug initiatives are less pronounced, these programs still play a critical role in prevention, rehabilitation, and support services.

The health-focused programs associated with social protection entail initiatives that can enhance psychological well-being and reduce financial stress, which are known to impact SHS. Moreover, programs integrating health with education include health promotion in schools, nutritional education programs, and first aid and personal care training. These initiatives can improve population perceptions of health by informing and equipping communities with the knowledge necessary to prevent diseases and maintain a healthy lifestyle.



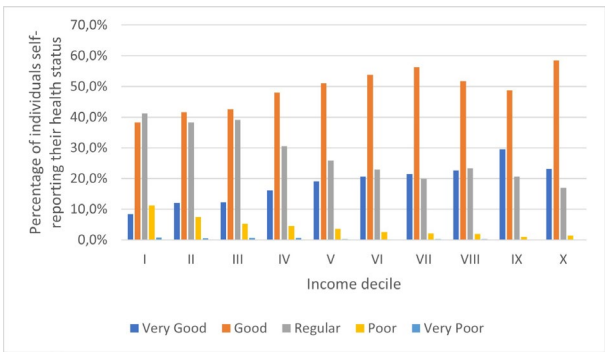
Note: women = 2166, N men = 2208. Mean subjective health status was higher among women (2.30, 95% CI [2.26, 2.34]) compared to men (2.14, 95% CI [2.11, 2.17]) ($F = 67.293, p < 0.001, df = 1, 6689$). Differences between age groups were also significant at $p < 0.05$ ($F = 59.599, p < 0.001, df = 6$), with mean health status increasing from 1.99 (95% CI [1.94, 2.03]) in the youngest group (18–28 years) to 2.76 (95% CI [2.65, 2.87]) in the oldest group (81+ years).

Fig. 2 Positive Self-reported Health Status by sex and age. Note: women = 2166, N men = 2208. Mean subjective health status was higher among women (2.30, 95% CI [2.26, 2.34]) compared to men (2.14, 95% CI [2.11, 2.17]) ($F = 67.293, p < 0.001, df = 1, 6689$). Differences between age groups were also significant at $p < 0.05$ ($F = 59.599, p < 0.001, df = 6$), with mean health status increasing from 1.99 (95% CI [1.94, 2.03]) in the youngest group (18–28 years) to 2.76 (95% CI [2.65, 2.87]) in the oldest group (81+ years)



Note: N = 4134. Mean subjective health status significantly differs across educational levels ($F = 138.815, p<0.001, df = 3, 4130$). Mean health status was highest among individuals with no education (2.73, 95% CI [2.64, 2.82]), followed by those with primary education (2.33, 95% CI [2.29, 2.37]), secondary education (2.08, 95% CI [2.05, 2.11]), and higher education (1.91, 95% CI [1.87, 1.95]).

Fig. 3 Self-reported health status by maximum educational level. Note: *N* = 4134. Mean subjective health status significantly differs across educational levels ($F = 138.815, p < 0.001, df = 3, 4130$). Mean health status was highest among individuals with no education (2.73, 95% CI [2.64, 2.82]), followed by those with primary education (2.33, 95% CI [2.29, 2.37]), secondary education (2.08, 95% CI [2.05, 2.11]), and higher education (1.91, 95% CI [1.87, 1.95])



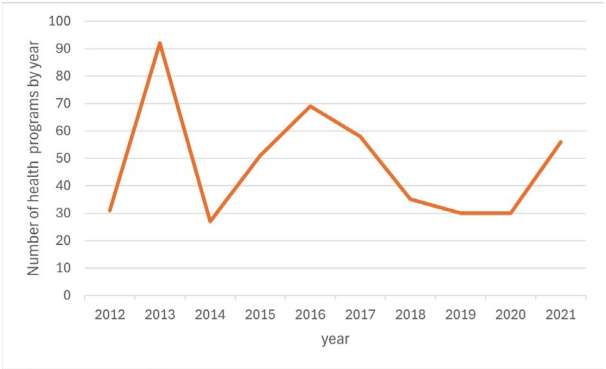
Note: N = 6391. Mean subjective health status significantly differs across income deciles ($F = 40.223, p < 0.001, df = 9, 6381$). 95% confidence intervals for the mean health status by income decile are as follows: 1st decile (2.57, 95% CI [2.50, 2.64]), 2nd decile (2.42, 95% CI [2.37, 2.47]), ..., 10th decile (1.97, 95% CI [1.89, 2.05]).

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Fig. 4 Self-reported Health Status by income decile. Note: *N* = 6391. Mean subjective health status significantly differs across income deciles ($F = 40.223, p < 0.001, df = 9, 6381$). 95% confidence intervals for the mean health status by income decile are as follows: 1st decile (2.57, 95% CI [2.50, 2.64]), 2nd decile (2.42, 95% CI [2.37, 2.47]), ..., 10th decile (1.97, 95% CI [1.89, 2.05])

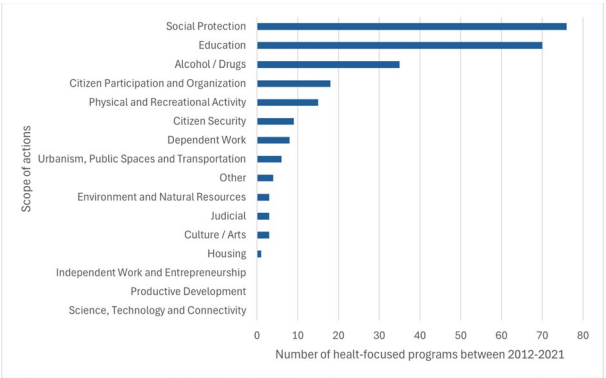
The mid-range sectors, such as citizen participation and organization and physical and recreational activity, show moderate levels of integration with health programs, illustrating a targeted but less comprehensive approach within these areas. However, such inclusion is notably infrequent in urbanism, public spaces and transportation, environment and natural resources, and judicial services. These areas, along with independent work and entrepreneurship, productive development, and science, technology, and connectivity, indicate a potential gap in the integration of health programs.

Additionally, we evaluated whether health-focused programs have a positive impact on people's income poverty and multidimensional poverty (measured in Chile across five dimensions: education, health, housing and environment, work, and social cohesion) since 2018, programs have reported on their contributions to reducing both multidimensional and income poverty. From 2018 to 2021, 83% (113 programs) reported contributing to reducing multidimensional poverty. In contrast, only one program reported efforts to reduce income poverty, providing free healthy food to vulnerable socioeconomic



Note: N=479 Health-focused programs, 2012-2021 .

Fig. 5 Health-focused programs in Chile by year. Note: N=479 Health-focused programs, 2012–2021



Note: N=456 Health-focused programs, 2012-2021

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Fig. 6 Health-focused programs by scopes of actions. Note: N=456 Health-focused programs, 2012–2021

groups in the Chilean population, and has been implemented systematically over time (The National Complementary Feeding Program (PNAC) nutrition program).

As anticipated, all health-focused programs positively impacted access to health services, and 85.5% noted a significant contribution to social cohesion and networks. However, only 8% of these programs contribute to reducing poverty related to housing and the environment, with efforts focused on controlling, monitoring, and regulating environmental factors that could impact the health of specific Chilean population groups. None of the health-focused programs positively impacted education (measured by attendance, school delays, and educational level) or work and social security (measured by access to occupation, retirement, and social security).

We also observed the contribution of health programs to various Sustainable Development Goals (SDGs) between 2017 and 2021 (Fig. 7). Good Health and Well-being (SDG 3) is the primary focus, with approximately 160 programs aligned. Other significant contributions include 'Zero Hunger' (SDG 2) and 'Sustainable Cities and Communities' (SDG 11), each involving nearly 100 programs. This distribution reflects the prioritization of health-related goals but also reveals a limited focus on SDGs like 'Climate Action' (SDG 13) and 'Life Below Water' (SDG 14), suggesting areas for future policy emphasis.

Discussion

In 2011, the World Health Organization published a bulletin titled "Social Determinants of Health" [18], stating that the health status of populations would not improve unless countries were capable of investing in and enhancing educational levels and training throughout people's lives, creating jobs with favorable working conditions, and providing a minimum income to meet basic needs for health and housing. A decade later, our analysis highlights social determinants' effect on the SHS of Chileans, revealing it to be a structural impact that persists over time. While SHS has progressively improved over time

at an aggregate level, when controlled for other socioeconomic and demographic aspects, the conclusion is that low income, low levels of education, and being retired or on a pension negatively predict the SHS today and 30 years ago.

The findings of this study must be interpreted within the broader socio-political and economic realities of Chile and Latin America. Chile, while often considered one of the region's most economically developed countries, is marked by deeply entrenched inequalities. The country's Gini index remains among the highest in Latin America, reflecting significant income disparities that directly affect health outcomes, including subjective health status [11].

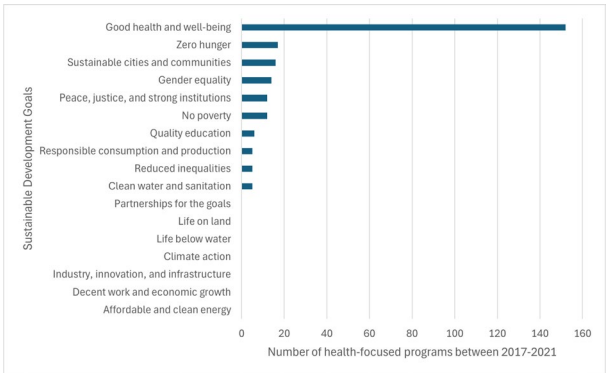
Public education in Chile has long been underfunded, creating a two-tiered system where private institutions offer significantly better resources and opportunities. This disparity limits social mobility and perpetuates cycles of inequality. In parallel, the privatized pension system, introduced during the military dictatorship, has left many older adults with insufficient income to meet their basic needs, negatively impacting their health and well-being [9].

These structural challenges are not unique to Chile but are emblematic of broader trends across Latin America. Studies have consistently shown that inequality in access to education and healthcare is a significant determinant of SHS throughout the region [6]. However, Chile's distinctive policy framework, characterized by privatized social services and a dual healthcare system, creates unique dynamics. For instance, the coexistence of a public health system (FONASA) serving the majority and a private system (ISAPRE) catering to higher-income groups amplifies disparities in perceived health outcomes.

The findings of this study align with these structural inequities. Individuals in lower income quintiles and those with lower educational attainment reported poorer SHS, reflecting the compounded disadvantages faced by these groups. Moreover, women, who often face additional barriers due to gendered labor market inequities and caregiving responsibilities, consistently reported worse SHS compared to men.

Regarding the effect of income on SHS, our results are consistent with numerous studies that declare income level a robust predictor of SHS throughout a person's life. A low socioeconomic status during childhood hurts the SHS reported in adulthood [19]. Similarly, moderate incomes are detrimental to the SHS in middle adulthood [20] and adults over 70 [21]. In Chile, wide gaps in health perception have been found between the socioeconomic groups of the highest and lowest deciles [9] and low SHS in the lowest income quintile [22]. Even in contexts of crisis and natural disasters that impact people's lives, those with low incomes exhibit worse levels of SHS [23, 24]. This indicates that an individual's income (and their household's) is a systematic and structural social determinant of people's health status.

The effect of education, measured in years of study, has also been documented as a factor explaining people's health status throughout their lives. The evidence indicates that, with higher formal education, individuals can make more efficient decisions regarding how to invest their time and energies, positively affecting their overall health status [25–27]. Education is also strongly associated with the existence of psychological, social, and financial resources that enable coping with unforeseen events affecting individual health and promoting self-care [26, 28, 29]. More educated individuals are better prepared to use new technologies, adjust to changes such as taking on a new job or losing it [17, 30, 31], and experience fewer



Note: N=244 Health-focused programs, 2017-2021).

Fig. 7 Health-focused programs declaring contributions to SDGs. Note: *N*=244 Health-focused programs, 2017–2021)

adverse effects on their physical and mental health when retiring early from the labor market [32] . In another line of research, education has been investigated as a mitigator of depressed health states in later stages of life, especially for current-generation women [25, 33].

Another predictive factor of the SHS is the employment situation of individuals. Our results show that retirement or pensioning harms SHS, controlling for time. These findings are consistent with other studies where being actively employed is associated with a positive SHS, while being physically disabled or unemployed correlates with a negative SHS [12, 34]. This can be explained by the significant reduction in income during the transition from worker to retiree [35]. Furthermore, evidence suggests that not having an active role in the labor market is associated with a loss of life's meaning, self-esteem, functionality, and socialization with others, leading to a diminished perception of health [36].

Other variables that showed a negative effect on SHS were being self-employed or being a homemaker, both understood as means to access income. The former may be associated with greater job instability at certain times, fluctuations in income levels, reduced access to financial aid, and increased exposure to financial stress during unforeseen events [17, 30, 37]. On the other hand, homemaking involves performing unpaid work associated with high physical and emotional overload and limited economic autonomy [27]. Both occupational categories negatively impact SHS; however, this effect disappears when controlling for time, indicating that individuals transition between various occupational options throughout their productive lives.

Being a parent seems to positively affect SHS, even when controlling income and educational level, but further exploration of this finding throughout the life cycle is necessary. It has been found that having one or more children positively affects SHS in older adults [21, 24], but it can represent an emotional and economic stressor in middle adulthood [38] in particular contexts such as the lockdown period during the COVID-19 pandemic [28], and especially for those with lower educational levels and incomes [20]. Furthermore, making distinctions according to the number and sex of children could provide greater clarity in implementing interventions that increase SHS in specific population groups. Additionally, evidence suggests that the economic, social, and psychological support provided by living as a couple positively affects SHS [27], especially in older adults [31, 39], we did not find support for this in our results. Future research could explore the impact that specific events such as widowhood or separation have on SHS depending on the life cycle and the existence or absence of additional support networks.

As expected, we found that age is a robust predictor of SHS. The positive perception of overall health status decreases with age [12, 20, 33, 40]; however, other variables are at play. Adults with chronic conditions experiencing low functionality present lower SHS than their healthy peers [21, 29], and unfavorable SHS has been found in adults of productive age engaged in poorly paid occupations [12]. Evidence also reports lower SHS and higher negative emotional susceptibility in older people during the pandemic for COVID-19 [41]. Interestingly, a national study reported high levels of SHS in socioeconomically vulnerable older adults aged between 65 and 90 years [10]. This leads us to explore other positive predictors of SHS beyond education and income as social determinants of health. Our hypothesis for this age group would be linked to the existence of support networks and social capital. Moreover, local evidence of SHS in youth populations is almost nonexistent, perhaps because with a high SHS, research focuses on priority groups, overlooking that there may be equally vulnerable subgroups among the young.

We also found that being female hurts SHS, which is consistent with numerous studies where men experience a better SHS than women, even when controlling for education and income [12, 24, 25, 33, 39]. This disparity is rooted in differences associated with biological sex and compounded by structural inequalities that disproportionately affect women. These include unequal distribution of caregiving responsibilities, labor market disadvantages, and disparities in access to resources, which can place additional burdens on women, particularly during middle and older adulthood, impacting their health perceptions.

Our findings reveal an intriguing relationship between sex and age. Between the ages of 61 and 70, SHS shows no significant differences between men and women, potentially reflecting a convergence in health experiences as both groups face similar life transitions, such as retirement. In the age group 71 to 80, men present a higher SHS than women, possibly due to gendered differences in chronic illness prevalence or access to healthcare services. Interestingly, in the segment of 81 years and older, women report better SHS than men. This could reflect resilience factors, such as stronger social networks and caregiving support, often observed among older women.

These results suggest that health promotion policies should consider the intersectionality of age and sex while also integrating a broader gender perspective. While this study specifically measured biological sex, future research could expand on these findings to explore the role of gendered social determinants, such as caregiving roles and labor market participation, in shaping SHS trajectories. Targeted interventions should address the

distinct vulnerabilities of women in early old age and support the well-being of men in advanced old age. Understanding the mechanisms underlying these associations, including social support structures, chronic disease patterns, and healthcare access, is crucial for informing evidence-based policies and programs.

The analysis based on health-focused programs highlights links between these initiatives and key SHS predictors such as social protection, education, and alcohol and drug programs. However, these relationships are observational and do not establish causal links. This integration suggests potential pathways for a holistic approach to health policy that could simultaneously address multiple social determinants of health.

The robust connection with social protection and education suggests that these areas are recognized as fundamental to improving health outcomes and effectively targeted by current policies. However, the limited interaction with sectors like science and technology, urbanism, public spaces, and environmental management points to a significant gap. These sectors are crucial for creating healthy living environments and can profoundly impact public health. The minimal integration observed suggests that current health policies may be missing opportunities to leverage improvements in these areas for broader health benefits. Some evidence provides positive inputs for people's quality of life using a research-policy perspective on urban environmental health and sustainability [42] and intelligent technologies applications for healthcare services [43].

Our findings indicate that health-focused programs declare to reduce multidimensional poverty, highlighting their potential role in addressing broader social determinants of health. However, the direct impact of these programs on SHS or income poverty remains an area for further investigation, as our analysis does not evaluate causal mechanisms.

Otherwise, the substantial contribution of those programs to reducing multidimensional poverty indicates that these initiatives play a critical role in addressing broader social determinants of health. However, while the singular focus on successfully enhanced access to health services and fostered social cohesion, their impact on education and employment-related housing and environment outcomes still needs to be improved. The lack of programs targeting these areas could contribute to persistent health disparities, particularly among the most vulnerable populations. The evidence supports the relevance of undertaking intersectional programs where health interacts with other life dimensions, promoting SHS beyond healthcare services [20, 28, 44]. Future initiatives could benefit from a deeper integration between health and educational sectors, extending beyond health promotion. Similarly, the relationship between health programs and improvements in occupational access and social security for retirees requires strengthening. This is particularly crucial given the demonstrated adverse impact of financial strain on SHS [45–47].

Regarding the distribution of health programs' contributions to various SDGs between 2017 and 2021, our analysis reveals a substantial emphasis on certain SDGs, reflecting Chile's strategic priorities in health-related social programs. The most pronounced focus is on 'Good Health and Well-being' (SDG 3), with contributions to 'Zero Hunger' and 'Sustainable Cities and Communities' reflecting an alignment with a holistic approach to health. While these efforts demonstrate a strategic prioritization of health-related SDGs, their specific impact on SHS remains to be explored in detail. Further research is needed to assess how these programs influence health perceptions across different population groups.

Contributions to "Zero Hunger" (SDG 2) and "Sustainable Cities and Communities" (SDG 11) are also significant, with each area nearly represented by 100 programs. These contributions reflect a holistic approach to health that incorporates food security and sustainable urban development as integral elements of health promotion. The alignment with "Zero Hunger" highlights the intersection of nutrition and health, recognizing that adequate nutrition is foundational to good health. Similarly, the focus on "Sustainable Cities and Communities" acknowledges the importance of healthy environments in shaping individuals' health outcomes.

Moderate engagement is observed in SDGs like "Gender Equality" (SDG 5) and "Peace, Justice, and Strong Institutions" (SDG 16), with contributions ranging from around 40 to 80 programs. While not as heavily emphasized, these areas are still crucial as they address underlying social determinants of health, such as gender disparities and the need for equitable and inclusive social systems, which directly influence individual and community health outcomes. Further research should consider interdisciplinary studies that examine the interplay between health outcomes and broader determinants such as gender equality and social justice. By investigating these connections, scholars can offer a more nuanced understanding of how holistic approaches to health and well-being might be achieved in line with the SDGs.

However, the analysis indicates a lesser focus on SDGs such as "Life Below Water" (SDG 14) and "Climate Action" (SDG 13), which have the fewest aligned programs. This suggests a narrower scope of engagement in environmental issues within the health programs. Although these SDGs are less directly related to

health, the emerging global health challenges associated with climate change and ecological degradation suggest that these areas require greater integration into health programming to address long-term health impacts comprehensively.

Policy recommendations

These recommendations address the immediate health needs and tackle the broader social and economic determinants of health, thereby improving the overall SHS of the population. Firstly, we recommend developing and expanding integrated programs that link health services with educational opportunities and occupational support. These should mainly target high-risk populations due to poor academic and employment outcomes. Examples include embedding health literacy programs in schools and workplaces and linking health services with vocational training and job placement services, mainly focusing on retirees and those in transitional employment.

Secondly, interventions should be designed to address the needs of the most vulnerable populations, such as low-income quintiles, retirees, and homemakers. These programs should aim to reduce financial strain through subsidies, direct financial support, or community-based programs that offer tailored health and social services. For instance, health subsidies could be specifically targeted to cover essential medications and preventive care for retirees and homemakers. Community centers could be established or expanded to provide free or low-cost health services, including mental health support and chronic disease management. Additionally, local health programs could partner with financial advisors to offer budget management and financial planning workshops for low-income families, enhancing their financial literacy and reducing stress.

Finally, while this study focuses on the socioeconomic determinants of SHS, such as income, education, and occupational stability, its findings underscore the need for a broader, multisectoral approach to health promotion. Future research could explore integrating health-focused programs with underrepresented SDGs, such as Climate Action (SDG 13) and Life Below Water (SDG 14). Addressing environmental health considerations, for instance, through community training on climate resilience or water conservation initiatives, could extend the impact of health policies beyond traditional domains. These strategies, though outside the immediate scope of this study, align with the growing recognition that environmental degradation disproportionately affects vulnerable populations, further compounding existing health inequalities.

Despite its contributions, this study has several limitations that should be acknowledged. First, data availability needs to restrict the inclusion of certain variables that could further enrich the analysis of SHS. Key factors such as territorial differences (rural versus urban, capital versus regions), the type and intensity of social capital, occupational characteristics, and lifestyle factors were not assessed. Additionally, the SHS measure relied on a single global question, which, while widely used, lacks the nuance of a multidimensional framework that could provide a more comprehensive understanding of individuals' health perceptions.

Although SHS is a robust predictor of health outcomes, it inherently reflects subjective perceptions influenced by cultural, psychological, and social norms. This variability may limit the comparability of SHS across diverse subgroups within the Chilean population. The study also does not account for intersectional dimensions, such as the effects of gender or ethnic discrimination. It does not consider contextual factors like the quality of healthcare services or acute stressors, which could significantly shape health perceptions. Lastly, the use of secondary survey data, while valuable, constrained the analysis of economic disparities to quintiles and deciles, avoiding more informative insights into income gaps. Moreover, the temporal scope of the study, which ends in 2019, does not capture the potential impact of the COVID-19 pandemic on SHS, an area warranting further investigation.

Conclusions

The findings of this study underscore the critical influence of demographic and socioeconomic factors on SHS in Chile, highlighting persistent structural inequalities. Over the past three decades, low income, limited educational attainment, and occupational status emerged as robust predictors of poor SHS, pointing to systemic barriers that require targeted policy interventions. This study also emphasizes the multidimensional nature of SHS, as it reflects the interplay of physical, mental, and social well-being shaped by broader societal determinants.

The analysis of health-focused programs revealed that the populations targeted by these initiatives often reflect the sociodemographic vulnerabilities identified in this study, such as low-income households, retirees, and individuals with limited educational attainment. This alignment underscores the importance of designing and implementing health policies directly addressing these structural inequities. By focusing on high-risk groups, these programs can mitigate disparities in SHS, though further integration with other social and economic sectors remains a critical area for improvement.

Although the study contributes valuable information, it also highlights limitations, including relying on aggregated survey data and excluding variables such as social capital and rural-urban disparities. Furthermore, this study used cross-sectional samples from different time periods instead of continuous panel data. Although the methodology ensures robust population-level insights, it limits the ability to track individual changes over time. Addressing these gaps in future research will improve our understanding of SHS and inform policies that are more inclusive and adaptive to the needs of diverse populations. Ultimately, this research reinforces the importance of evidence-based strategies in promoting equitable health outcomes and improving the overall SHS in Chile.

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Authors' contributions
M.B led the study's conception, design, data analysis, and interpretation, as well as the draft and revision of the manuscript. MB also managed communication with the journal and ensured compliance with transparency and reproducibility standards. J.R, J.O, M.S and C.P contributed to data analysis, interpretation, literature review, and methodology development. They reviewed and revised the manuscript and approved the final version, ensuring accountability for their contributions.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
This research used secondary open Access only. All data are available at https://www.worldvaluessurvey.org/wvs.jsp and https://programassociales.ministeriodesarrollosocial.gob.cl/programas.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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